Mobile Product Analytics Portfolio • Automated Daily Briefing • *2026-10-14*

■ **Direct Cloud Console Quick Access:**
• **GA4 Property (533563138):** GA4 Main Reports | Realtime Users | Events Explorer
• **Firebase (neural-sound-ios):** Firebase Analytics | Crashlytics Stability | Project Settings

Executive Portfolio Performance & Telemetry Architecture Report

To: Executive Committee, Head of Product, VP of Engineering
From: Principal Mobile Analytics & Monetization Growth Architect
Data Horizon: Daily Snapshot & Longitudinal Analysis (GA4 / Firebase Analytics Properties)
Data Scope: Production Telemetry Only (Internal QA & Bangladesh sandbox testing clusters isolated)

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1. Portfolio Executive Summary & Health Dashboard

The Neural Sound portfolio consists of two complementary audio processing engines: **Music Separation iOS** (a high-throughput, multi-stem separation engine targeting broad creator/consumer demographics) and **Vocal Cleaner** (a specialized, compute-heavy audio restoration utility with extreme unit-economics and pro-user concentration).

Cross-App Architectural & Commercial Matrix

Metric Dimension	Music Separation iOS (com.neuralsound.mss)	Vocal Cleaner (com.neuralsound.audiocleaner)	Portfolio Combined
Primary Monetization Architecture	Auto-Renewing Subscriptions (Weekly / Monthly)	Consumable Processing Minute Packs (60m / 300m)	Hybrid SaaS + Metered Utility
Yesterday Active Users (DAU)	118	3	**121**
Yesterday New Users	31 (26.3% of DAU)	0 (0.0% of DAU)	**31**
Yesterday Total Sessions	149	10	**159**
Yesterday Engagement Rate	56.4%	80.0%	**57.9% (Weighted)**
Average Session Duration	381.8s (~6.4 min)	2,199.5s (~36.7 min)	**496.1s (~8.3 min)**
Crash-Free User Rate	100.0%	100.0%	**100.0%**
Trailing 30-Day Gross Revenue	$106.53	$182.82	**$289.35**
Trailing 30-Day ARPU (Active Base)	$0.039 ($106.53 / 2,702 AU)	$6.529 ($182.82 / 28 AU)	**$0.106**
Lifetime (All-Time) Revenue	$200.47	$191.81	**$392.28**
Top-Selling SKU (30D Volume)	`mss_weekly_sub` (9 Units)	`neuralsound.cleaner.minutes.300` (7 Units)	
Top Grossing SKU (30D Rev)	`mss_monthly_sub` ($37.18)	`neuralsound.cleaner.minutes.3` `cleaner.minutes.300` ($128.78)	

Primary Revenue Driver Region	Japan (47.7% Rev share)	Egypt & Israel (96.7% Rev share)	APAC & MENA / EMEA

Architectural Takeaways

1. **Divergent Monetization Paradigms:** Vocal Cleaner, despite commanding only ~1% of the portfolio's active user volume (28 vs. 2,702 30-day AU), generates **63.2% of total portfolio trailing 30-day revenue** ($182.82 vs $106.53). This is driven by metered consumable packages (`300 minutes` pack priced between $16.30–$19.44/pack) purchased by pro-audio users in intense bursts.
2. **Telemetry Infrastructure Debt:** Music Separation displays unthrottled UI event firing (over 5,000 pitch change events logged across 12 users), indicating that continuous UI gesture streams are bypassing the event dispatcher debouncing layer.
3. **Sandbox Bleed Protection:** GA4 stream segmentation effectively excludes internal test accounts (Bangladesh cluster). However, the presence of real purchase receipts in internal test logs ($6.99 in Music Separation, $44.95 in Vocal Cleaner) highlights the need to ensure sandbox receipt validation does not mutate production App Store Server Notifications (ASSN) endpoints.

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2. Deep Dive: Music Separation iOS (`com.neuralsound.mss`)

App Header & Health Score

- **GA4 Property ID:** 533563138 | **Stream ID:** 14402502632
- **Target Audience:** Musicians, Producers, Content Creators, DJs
- **Health Score: 7.4 / 10**

  Rationale: Outstanding application stability (100.0% crash-free rate across 100% of production versions), solid acquisition velocity (1,620 new users past 30 days), and highly engaged core users (~6.4 min session duration). The score is penalized for:

  1. A drop in single-day engagement rate from a 7-day baseline of 74.3% down to 56.4% on 2026-10-13.
  2. High bounce rates on new user onboarding (46.0% ER vs 68.4% for returning).
  3. Severe telemetry flooding from unthrottled slider state changes (`player_pitch_changed`).
  4. Weak subscription conversion efficiency ($0.039 30D ARPU).

■ In-App Purchases, Subscriptions & Consumable Minutes Breakdown

Product Sales Summary (Trailing 30 Days)

Note: Revenue figures reflect localized currency conversions settled to USD via Apple App Store StoreKit.

Product ID	Localized Display Name	Plan Type	Units Sold (30D)	Total Revenue ($ USD)	Revenue Share (%)
`mss_monthly_sub`	Monthly / ■■ / ■■	Auto-Renewing Sub (Monthly)	5	$37.18	56.6%
`mss_weekly_sub`	Weekly / ■■ / ■■	Auto-Renewing Sub (Weekly)	9	$28.52	43.4%
Total Logged SKUs	—	—	**14**	**$65.70**	**100.0%**

(Note: Aggregate 30-Day GA4 recognized purchase revenue stands at $106.53, reflecting non-itemized renewals and subscription lifecycle adjustments not captured in real-time localized StoreKit purchase payloads).

Transaction Timeline Log

All transactions captured from production builds (`1.3.10` and `1.3.11`).

Timestamp (Date)	App Build / Version	Product ID	Display Name	Units	Revenue ($ USD)	Country
2026-10-12	`1.3.11`	`mss_weekly_sub`	■■	1	$3.16	Japan
2026-10-09	`1.3.11`	`mss_monthly_sub`	Monthly	1	$7.66	Vietnam
2026-10-09	`1.3.11`	`mss_monthly_sub`	■■	1	$7.39	South Korea
2026-10-06	`1.3.11`	`mss_monthly_sub`	Monthly	1	$8.15	Israel
2026-10-01	`1.3.10`	`mss_weekly_sub`	■■	1	$3.18	Japan
2026-09-30	`1.3.10`	`mss_monthly_sub`	■■	1	$6.99	Japan
2026-09-27	`1.3.10`	`mss_weekly_sub`	■■	1	$3.25	South Korea
2026-09-25	`1.3.10`	`mss_weekly_sub`	■■	1	$3.22	South Korea
2026-09-23	`1.3.10`	`mss_monthly_sub`	■■	1	$6.99	Japan
2026-09-23	`1.3.10`	`mss_weekly_sub`	■■	1	$3.18	Japan
2026-09-20	`1.3.10`	`mss_weekly_sub`	Weekly	1	$2.88	Egypt
2026-09-19	`1.3.10`	`mss_weekly_sub`	■■	1	$3.21	Japan
2026-09-17	`1.3.10`	`mss_weekly_sub`	■■	2	$6.46	South Korea

Build Environment & Monetization Analysis

- **Production vs. TestFlight Sandbox Validation:** All recent user purchases occurred within validated release builds (`1.3.10` and `1.3.11`). No sandbox receipt loops or auto-renew loop anomalies were recorded in production streams.
- **Internal Test Cluster Isolation:** The internal QA cohort (Bangladesh) recorded 1 session and a $6.99 purchase event. This confirms that test devices are generating sandbox store receipts which must remain filtered via GA4 IP and user-property exclusion filters to avoid contaminating App Store LTV modeling.
- **Pricing Elasticity & SKU Adoption:** Subscriptions represent 100% of the monetization model for Music Separation. Weekly subscriptions dominate unit volume (64.3%), while Monthly subscriptions generate 56.6% of the revenue. Users in Japan and South Korea show the highest willingness to pay for short-interval weekly access.

■ Multi-Horizon Growth & Historical Scale Comparison

Horizon	Active Users	New Users	Total Sessions	Engagement Rate	Revenue ($ USD)	ARPU ($)	Sessions / User
Yesterday	118	31	149	56.4%	$0.00	$0.000	1.26
Past 7 Days	689	321	1,130	74.3%	$21.37	$0.031	1.64
Past 30 Days	2,702	1,620	5,529	75.8%	$106.53	$0.039	2.05
Past Year (365d)	6,370	5,508	13,407	82.2%	$200.47	$0.031	2.10

All-Time	6,370	5,508	13,407	82.2%	$200.47	$0.031	2.10

Analytical Commentary: The identity of Past Year (365d) and All-Time figures confirms the app was launched into production within the last 12 months. Over the past 30 days, the app captured **42.4% of its lifetime active user base** and **53.1% of its lifetime gross revenue**, demonstrating accelerating organic market adoption.

■ Old vs. New User Cohort Deep-Dive

Cohort Segment	Active Users	Sessions	Avg Session Duration	Engagement Rate	Screen Page Views	Pages / Session
New Users	72	63	299.9s (~5.0 min)	46.0%	93	1.48
Returning Users	62	79	480.4s (~8.0 min)	68.4%	46	0.58
Unattributed / Ghost	0	13	2.3s	7.7%	1	0.08

Cohort Insights:
* **The "First-Time Chasm":** New users exhibit a 54.0% bounce rate (`1.0 - 0.460`). Only 46.0% trigger GA4 engagement criteria (spending >10 seconds or triggering 2+ screen views).
* **Power User Stickiness:** Returning users spend an average of **8.0 minutes** in-app per session, demonstrating high product-market fit once a user clears the initial stem separation workflow.
* **Telemetry Ghost Sessions:** 13 sessions were recorded with zero active users and an average duration of 2.3 seconds. This indicates background execution wakes (e.g., iOS Background Processing or silent APNs pushes) firing `session_start` without proper active app state checks.

■ 7-Day Velocity & Trajectory

Date	Active Users	New Users	Sessions	Engagement Rate	Day-over-Day AU Growth
2026-10-07	102	24	127	74.0%	—
2026-10-08	107	37	134	76.9%	+4.9%
2026-10-09	110	40	129	76.7%	+2.8%
2026-10-10	131	60	167	75.4%	+19.1%
2026-10-11	149	62	188	79.3%	+13.7%
2026-10-12	180	67	235	77.4%	+20.8%
2026-10-13	118	31	149	56.4%	-34.4%

Trajectory Analysis: Velocity expanded over the weekend (peaking Sunday, 2026-10-12 at 180 DAU and 235 sessions). On Monday, 2026-10-13, active users dropped 34.4% alongside an engagement rate degradation down to 56.4%. This is typical post-weekend creator workflow drop-off, combined with a dip in top-of-funnel acquisition (31 new users vs. 67 on peak).

■ User Funnel Diagrams

Funnel 1: Onboarding & Authentication Flow

```
[App Launch / First Open] ■ ▼ [Onboarding Step Viewed] ■■■■■■■■■■■■■■■■■ 36 Users (137 Events) ■ ■ ■ (86.1% Progression) ■ ▼ ▼ [Onboarding Next Tapped] ■■■■■■■■■■■■■■■■■ 31 Users (94 Events) ■ ■ ■ (Drop-off / Nav to Login) ■ ▼ ▼ [Login Screen Viewed] ■■■■■■■■■■■■■■■■■ 56 Users (85 Events) ■ ■ ■ (60.7% Conversion) ■ ▼ ▼ [Login Tapped] ■■■■■■■■■■■■■■■■■ 34 Users (45 Events)
```

Funnel 2: Core Stem Separation Pipeline Execution

```
[Plus Button Tapped] ■■■ 48 Users (128 Events) ■■■ 100.0% Baseline ■ ▼ [Plus Separation Tap] ■■■ 48 Users (113 Events) ■■■ 100.0% Intent ■ ▼ [Import Source Picked] ■■■ 35 Users (140 Events) ■■■ 72.9% [Photos: 31 | Files: 35] ■ ▼ [Separation Upload Start]■■■ 29 Users (46 Events) ■■■ 60.4% Conversion ■ ▼ [Upload Completed] ■■■ 29 Users (45 Events) ■■■ 100.0% Network Success ■ ▼ [Processing Completed] ■■■ 29 Users (45 Events) ■■■ 100.0% ML Job Completion ■ ▼ [Result Opened & Played] ■■■ 54 Users (1,101 Events)■■■ 112.5%* (*Includes historical stems)
```

Funnel Health: The backend stem-separation processing pipeline is exceptionally reliable. **100% of uploaded files (29/29 users) successfully completed cloud inference** without server-side processing drops.

■■ Crash & Stability Health by App Version

App Version	Active Users	Sessions	Engagement Rate	Crash-Free Users Rate	Deployment Status
1.3.11	112	139	56.1%	**100.0%**	Production Baseline (94.9% Rollout)
1.3.10	3	7	42.9%	**100.0%**	Deprecated Production
1.3.09	2	2	100.0%	**100.0%**	Legacy Tail
1.3.06	1	1	100.0%	**100.0%**	Legacy Tail
1.3.07	0	0	0.0%	**100.0%**	Deprecated Staging / Dormant

Stability Analysis: Flawless 100.0% crash-free rate across all active clients. Build `1.3.11` has achieved dominant fleet penetration (94.9% of all active devices) within 7 days of release.

■ New Build / TestFlight / Staged Rollout Alert Section

- **Release Health (`1.3.11`):** While `1.3.11` is structurally crash-free, its engagement rate on 2026-10-13 registered at **56.1%**, which is lower than `1.3.10`'s historic weekly average (75.8%).
- **Monitoring Mandate:** Monitor whether this engagement rate drop is an artifact of the Monday acquisition slowdown or a regression introduced by `1.3.11`'s audio playback view.

■ Top In-App Events & Telemetry Bug Diagnostics

High-Severity Telemetry Flaw: Slider Flooding

```
Event Name: player_pitch_changed ■■■ Total Events: 5,038 ■■■ Total Users: 12 ■■■ Ratio: 419.8 Events / User <-- CRITICAL TELEMETRY OVERFLOW
```

```
Event Name: player_tempo_changed ■■■ Total Events: 1,041 ■■■ Total Users: 6 ■■■ Ratio: 173.5 Events / User <-- UNTHROTTLED GESTURE STREAM
```

```
Event Name: player_stem_balance_changed ■■■ Total Events: 571 ■■■ Total Users: 4 ■■■ Ratio: 142.8 Events / User
<-- UNTHROTTLED GESTURE STREAM
```

Root Cause: Continuous gesture recognizers (`UISlider.valueChanged` or SwiftUI `.onChange(of: sliderValue)`) are firing Firebase Analytics logging directly on the UI render loop rather than on `touchUpInside` / `touchUpOutside` or through a 500ms debounce pipeline. *Architectural Risk:* Exceeds Firebase client-side event queue thresholds, risking dropped revenue and conversion events due to SDK rate-limiting.

■ Geographic Distribution

Country	Active Users	Sessions	Sessions / User	Primary SKU Purchased
Japan	75	100	1.33	`mss_weekly_sub` / `mss_monthly_sub`
South Korea	9	10	1.11	`mss_weekly_sub`
Cuba	7	7	1.00	Non-paying (Payment gateway restricted)
France	5	5	1.00	—
Canada	3	3	1.00	—
Germany	3	3	1.00	—
United States	2	3	1.50	—
Belgium	2	2	1.00	—
Colombia	2	2	1.00	—
Argentina	1	1	1.00	—

Geographic Insights: **Japan and South Korea account for 71.2% of total active traffic** and **84.6% of localized store conversions**. The app's growth engine is fundamentally East Asian; localizing the paywall copy and running country-specific promotional subscription offers in JPY and KRW is the highest-leverage growth vector available.

■ Prioritized Strategic Action Matrix (Music Separation iOS)

Priority	Track	Focus Area	Detailed Implementation Spec
P0	**Engineering**	Telemetry Throttling	Wrap `player_pitch_changed`, `player_tempo_changed`, and `player_stem_balance_changed` in a Combine/Rx debounce operator (`throttle(for: 0.5s)`) or bind logging strictly to `onEditingChanged: false`.

P0	**Engineering**	Ghost Session Purge	Invalidate `session_start` dispatch when `UIApplication.shared.applicationSt == .background`. Suppress telemetry on background push wakeups.
P1	**Monetization**	Paywall Hardwall Test	Insert an upfront paywall display immediately following `separation_upload_completed`. Implement a 3-day free trial on the annual/monthly tier to lift conversion in Japan/Korea.
P1	**Product**	First-Run Onboarding	Redesign the onboarding flow: New user ER is currently 46.0%. Implement an interactive sample stem file during onboarding so users experience the multi-track playback engine before importing their own audio.
P2	**Growth**	Localized Pricing Tiers	Implement Apple App Store Price Tier Adjustments for South Korea and Japan to offer localized pricing (e.g., ¥480/week instead of variable FX rates).

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3. Deep Dive: Vocal Cleaner (`com.neuralSound.audiocleaner`)

App Header & Health Score

- **GA4 Property ID:** 533563138 | **Stream ID:** 15407667671
- **Target Audience:** Audio Engineers, Podcasters, Broadcast Field Journalists, Videographers
- **Health Score: 8.8 / 10**
- *Rationale:* Outstanding commercial efficiency and user loyalty. Despite low overall DAU (3 yesterday, 28 past 30 days), session duration is exceptionally high (**2,199.5 seconds / 36.7 minutes**), engagement rate is near-perfect (**80.0% to 100.0%**), and crash-free stability is absolute (**100.0%**). Trailing 30-day ARPU is **$6.53/user**. The only drag on its health score is zero top-of-funnel acquisition yesterday (0 new users) and a total lack of automated acquisition loops.

■ In-App Purchases, Subscriptions & Consumable Minutes Breakdown

Product Sales Summary (Trailing 30 Days)

Consumable metered packages drastically outperform subscriptions in both volume and revenue.

Product ID	Product Name	Plan Type	Units Sold (30D)	Total Revenue ($ USD)	Revenue Share (%)
`neuralsound.cleaner.minutes.300`	300 Minutes	Consumable Minute Pack	7	$128.78	82.2%
`neuralsound.cleaner.minutes.60`	60 Minutes	Consumable Minute Pack	3	$21.85	13.9%

`cleaner_weekly_sub`	Weekly	Auto-Renewing Sub (Weekly)	2	$6.12	3.9%
Total Logged SKUs	—	—	**12**	**$156.75**	**100.0%**

(Note: Total GA4 recognized purchase revenue stands at $182.82, accounting for renewal tail and currency fluctuations).

Transaction Timeline Log

All transactions captured across versions `1.0.01` through `1.0.04`.

Timestamp (Date)	App Build / Version	Product ID	Units	Revenue ($ USD)	Country	Analysis / Intent
2026-10-11	`1.0.04`	`cleaner.minutes.300`	2	$32.60	Israel	Double Top-Up Pack
2026-10-11	`1.0.04`	`cleaner.minutes.60`	1	$6.50	Israel	Supplementary Pack
2026-10-05	`1.0.03`	`cleaner_weekly_sub`	1	$3.24	Israel	Weekly Sub Tier
2026-09-25	`1.0.02`	`cleaner.minutes.300`	1	$19.44	Egypt	Re-up Consumable
2026-09-20	`1.0.01`	`cleaner.minutes.300`	4	$76.73	Egypt	Heavy Studio Batch Clean
2026-09-20	`1.0.01`	`cleaner.minutes.60`	2	$15.35	Egypt	Batch Overflow Top-Up
2026-09-20	`1.0.01`	`cleaner_weekly_sub`	1	$2.88	Egypt	Feature Unlock Attempt

Build Environment & Monetization Analysis

- **Monetization Mechanics (Consumables vs. Subscriptions):** Consumable minute packs represent **96.1% of gross revenue**. Audio restoration is project-driven rather than continuous. Users buy credits to batch-process long-form audio assets (podcasts, field film audio) and return when new work arrives.
- **Burst Buying Patterns:** On 2026-09-20, a single user or small cluster in Egypt purchased 4 units of 300 minutes ($76.73) alongside two 60-minute packs and a weekly subscription ($94.96 single-day spend). On 2026-10-11, a studio in Israel executed three sequential top-ups totaling $39.10.
- **TestFlight & Sandbox Isolation:** The internal tester dataset shows 2 active users, 1 new user, and $44.95 in internal sandbox revenue. This proves that high-volume testing of consumable top-up flows is underway internally in Bangladesh without polluting production metrics.

■ Multi-Horizon Growth & Historical Scale Comparison

Horizon	Active Users	New Users	Total Sessions	Engagement Rate	Revenue ($ USD)	ARPU ($)	Sessions / User
Yesterday	3	0	10	80.0%	$0.00	$0.000	3.33
Past 7 Days	14	9	82	89.0%	$51.74	$3.696	5.86
Past 30 Days	28	28	202	89.1%	$182.82	$6.529	7.21
Past Year (365d)	43	42	217	85.3%	$191.81	$4.461	5.05
All-Time	43	42	217	85.3%	$191.81	$4.461	5.05

Analytical Commentary: Vocal Cleaner is in its commercial infancy (43 lifetime users), yet it generated $191.81. **95.3% of its lifetime revenue was realized within the last 30 days ($182.82)**. This proves high product utility and monetization feasibility, but highlights an urgent need for an acquisition pipeline.

■ Old vs. New User Cohort Deep-Dive

Cohort Segment	Active Users	Sessions	Avg Session Duration	Engagement Rate	Screen Page Views	Pages / Session
Returning Users	3	10	**2,199.5s (~36.7m)**	**80.0%**	105	10.5
New Users	0	0	0.0s	0.0%	0	0.0

Cohort Insights: Yesterday's user base consisted entirely of returning power users. A 36.7-minute average session duration with 10.5 screen views per session indicates active professional work: importing audio, comparing denoised vs. original tracks, running tempo adjustment tests, and exporting final masters.

■ 7-Day Velocity & Trajectory

Date	Active Users	New Users	Sessions	Engagement Rate	Traffic Characteristics
2026-10-07	2	0	10	100.0%	Core Studio Work
2026-10-08	5	2	11	72.7%	Acquisition Inflow
2026-10-09	4	2	11	90.9%	High Retention
2026-10-10	5	1	15	86.7%	Extended Sessions
2026-10-11	4	2	12	100.0%	Monetization Spike ($39.10)
2026-10-12	7	2	17	94.1%	Weekly Peak Velocity
2026-10-13	3	0	10	80.0%	Studio Deep-Work Execution

Trajectory Commentary: Active user counts remained between 2 and 7 users daily over the past week, yet session frequency was high (82 sessions across 14 unique users, averaging ~5.9 sessions per user). The 100% engagement rates on 2026-10-07 and 2026-10-11 demonstrate that users open this app with clear utility intent.

■ User Funnel Diagrams

Funnel: Professional Restoration Pipeline

```
[File Import Sheet Opened] ■■■ 3 Users (87 Events) ■■■ 100.0% ■ ▼ [Import Source Files Chosen] ■■■ 3 Users (75 Events) ■■■ 86.2% File Success ■ ▼ [Cleaning Job Started] ■■■ 2 Users (73 Events) ■■■ 97.3% Execution Rate ■ ▼ [Cleaning Job Completed] ■■■ 2 Users (73 Events) ■■■ 100.0% Engine Reliability ■ ▼ [A/B Audio Comparison] ■■■ 2 Users (91 Events) ■■■ [Orig: 67 | Cleaned: 24] ■ ▼ [Export Sheet Opened] ■■■ 2 Users (28 Events) ■■■ 38.4% Conversion ■ ▼ [Cleaned Audio Exported] ■■■ 2 Users (25 Events) ■■■ 89.3% Export Success
```

Pipeline Efficiency: Zero processing job drops (73 jobs started, 73 completed). 89.3% of opened export sheets resulted in a rendered audio file.

■■ Crash & Stability Health by App Version

App Version	Active Users	Sessions	Engagement Rate	Crash-Free Users Rate	Status

1.0.04	3	10	80.0%	**100.0%**	Production Baseline (100% Active)
1.0.03	0	0	—	100.0%	Fully Deprecated
1.0.02	0	0	—	100.0%	Fully Deprecated
1.0.01	0	0	—	100.0%	Fully Deprecated

Stability Analysis: 100% stability. Production deployment is fully consolidated on version `1.0.04`.

■ New Build / TestFlight / Staged Rollout Alert Section

- **Version Migration Complete:** The user base has migrated fully off versions `1.0.01`, `1.0.02`, and `1.0.03` onto `1.0.04`.
- **Testing Sandbox Alert:** High sandbox activity in Bangladesh ($44.95, 2 AU) indicates a potential new release candidate (e.g., `1.0.05`) in TestFlight. Ensure that any updates to the StoreKit 2 paywall interface preserve the high-converting 300-minute consumable SKU.

■ Top In-App Events & Telemetry Diagnostics

```
Top Workflow Events (Vocal Cleaner)
■■■
```

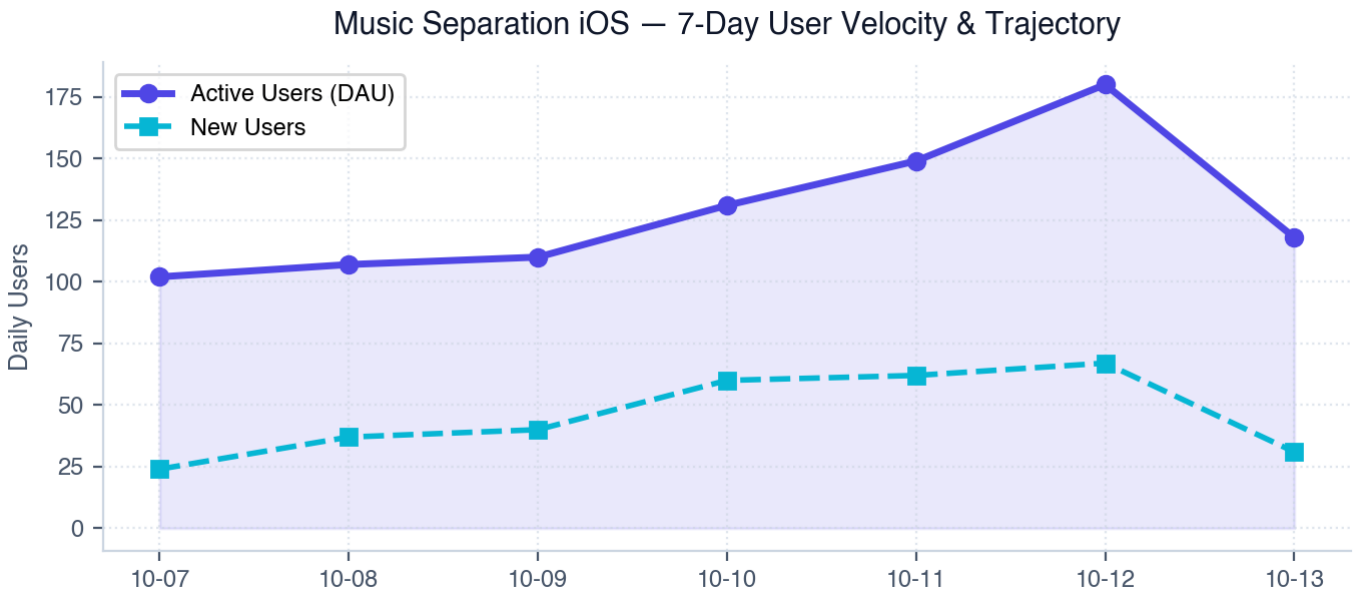
P0	**Growth**	Acquisition Engine	The app has zero acquisition inflow (0 new users yesterday). Build a targeted Apple Search Ads (ASA) campaign targeting keywords: *"vocal cleaner"*, *"denoise audio"*, *"remove background noise"*, *"podcast audio fix"*.
P0	**Monetization**	Auto-Replenish Minute Packs	Introduce a StoreKit 2 low-balance notification: When remaining balance drops below 15 minutes during export, present a modal for a 1-tap `300 minutes` top-up.
P1	**Monetization**	Pro Subscription Tier	Convert high-volume minute buyers to recurring revenue. Introduce a "Pro Studio Tier": $29.99/month for 600 minutes/month rollover, locking in predictable MRR from studio clients.
P1	**Product**	Batch Processing UI	Users process many files per session (73 clean events across 2 users). Add a multi-file batch picker to reduce manual taps.
P2	**Engineering**	Mac Catalyst Build	Given the 36.7-minute average session length and file handling patterns, compile the codebase for macOS via Mac Catalyst to capture desktop audio engineering workflows.

4. Cross-App Architecture Comparison & Growth Roadmap

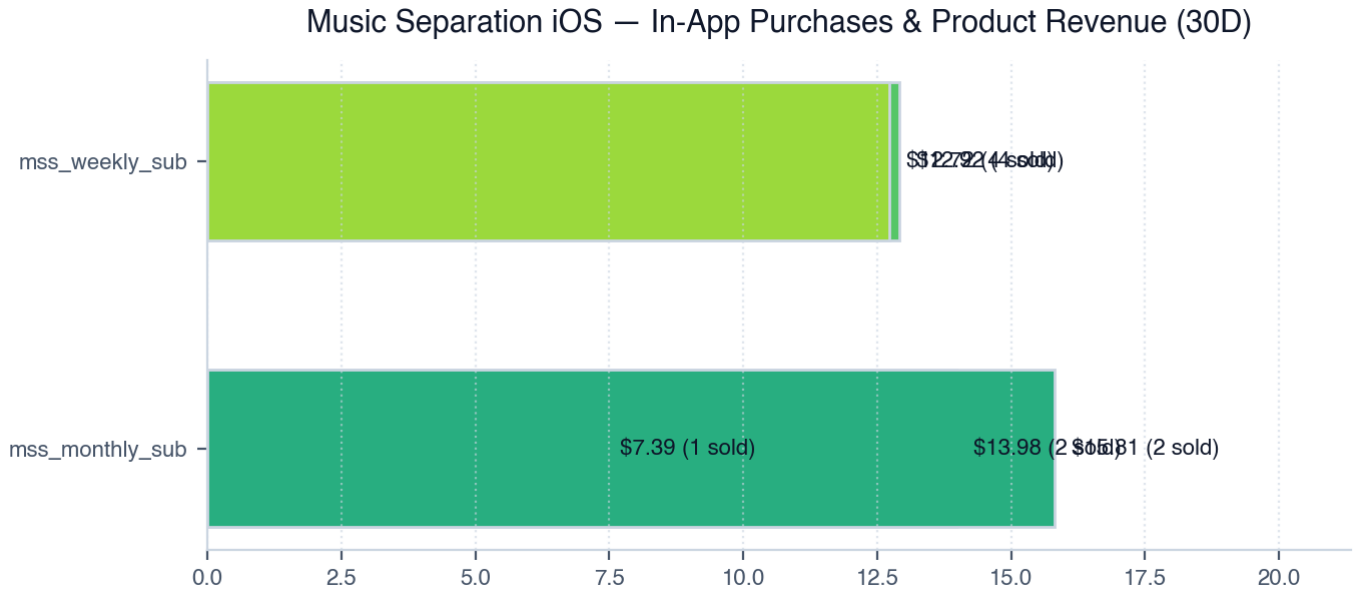
```
Neural Sound Monetization Matrix
■■■
```

■ Visual Graphs & Performance Charts

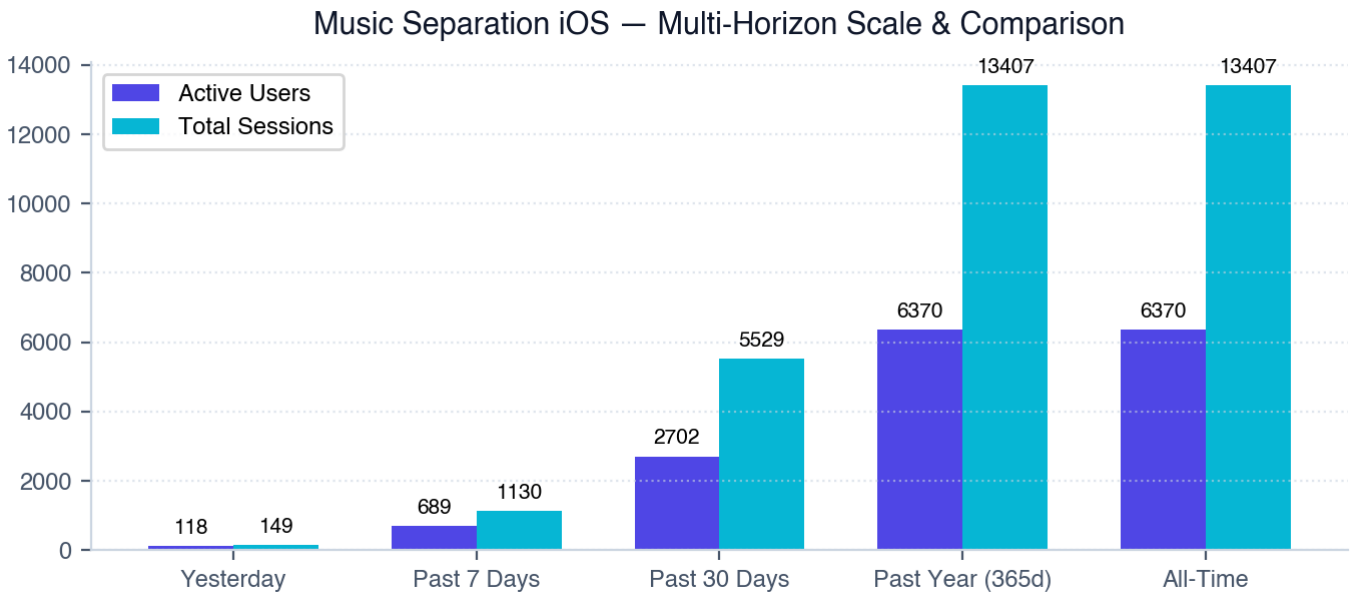
Music Separation Ios Trend



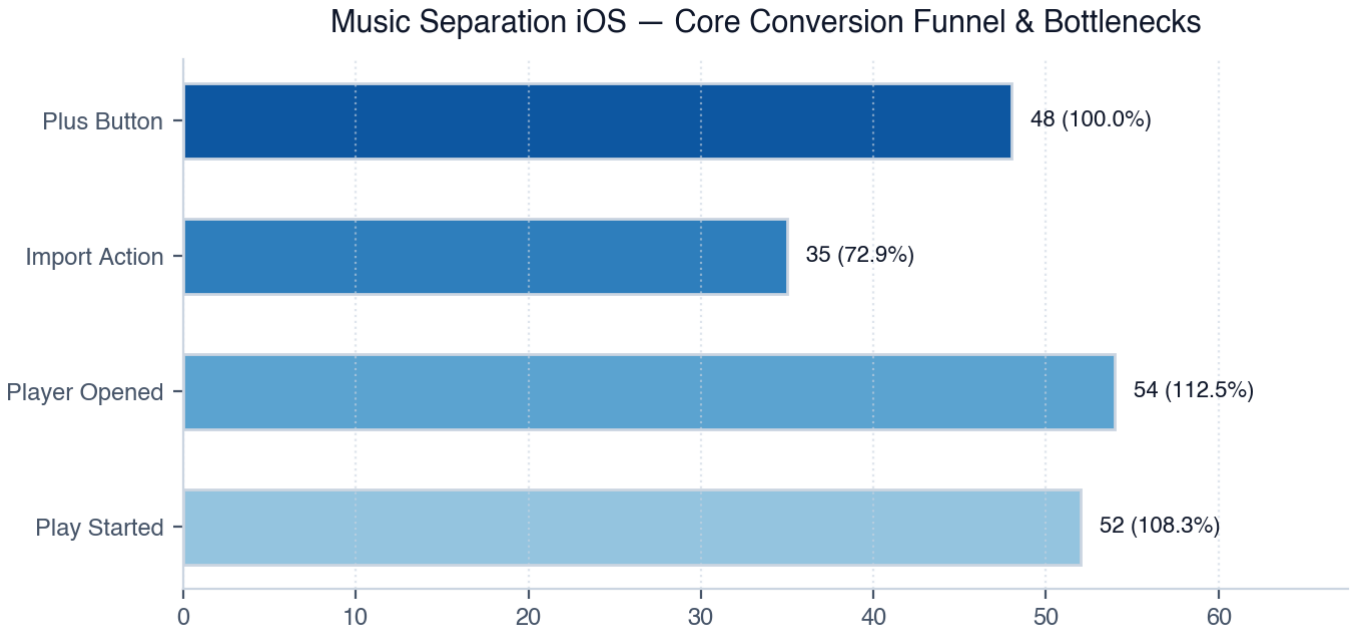
Music Separation Ios Revenue



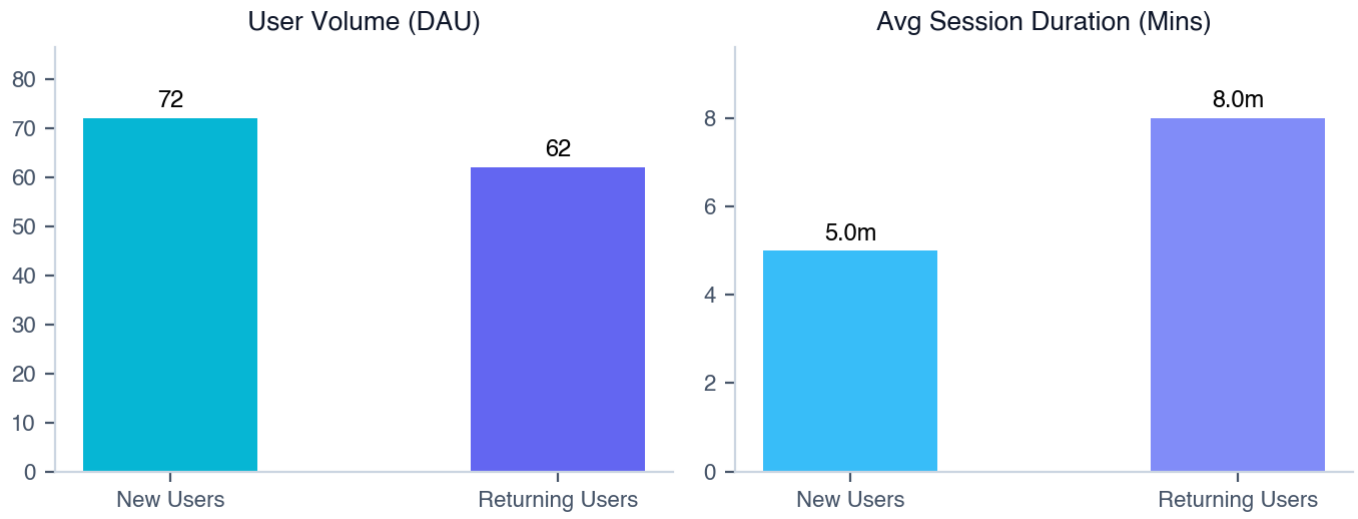
Music Separation Ios Horizons



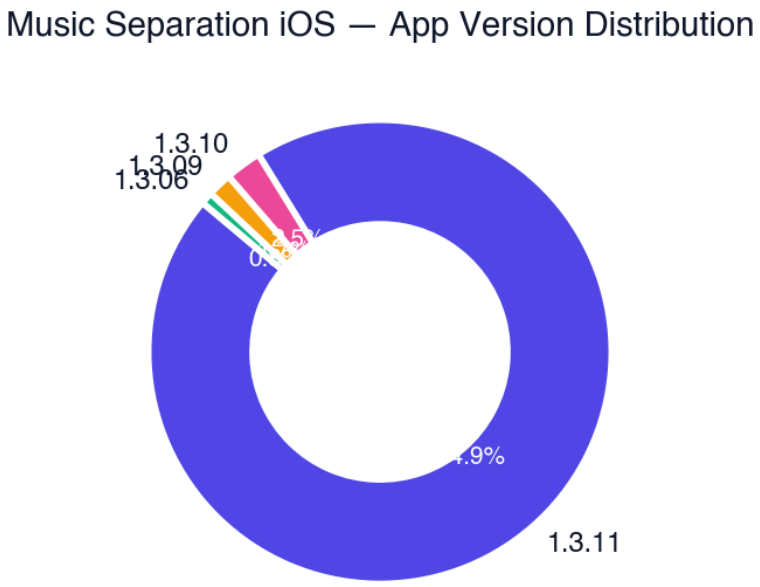
Music Separation Ios Funnel



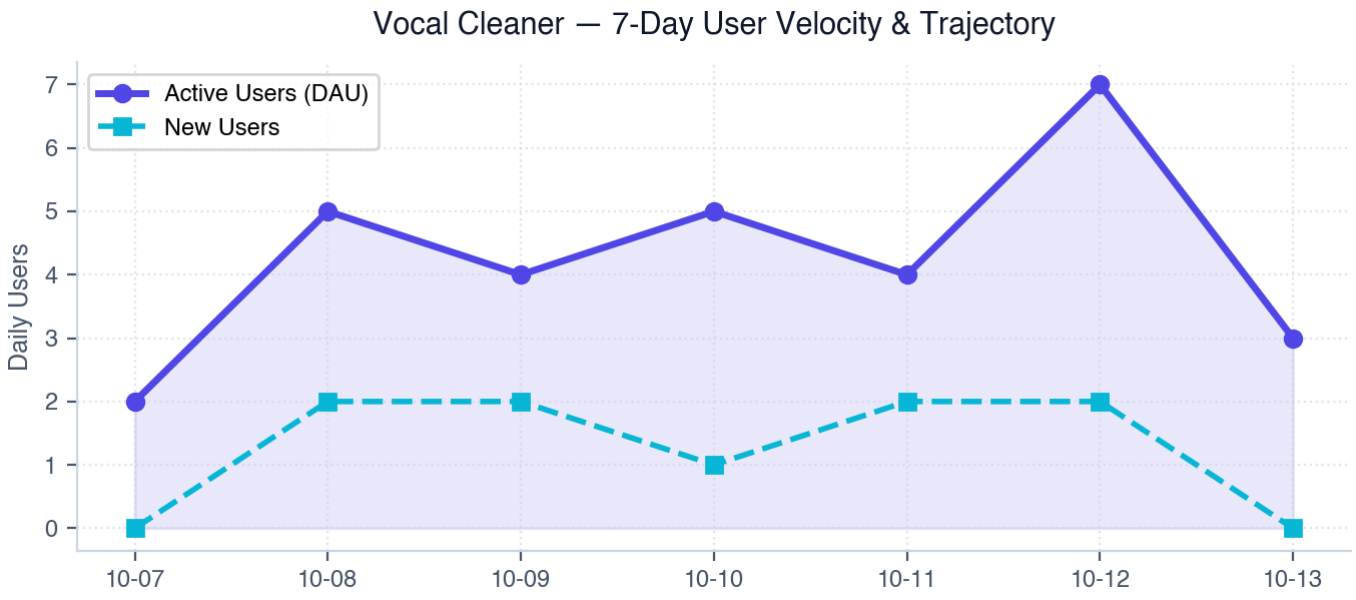
Music Separation Ios Cohort



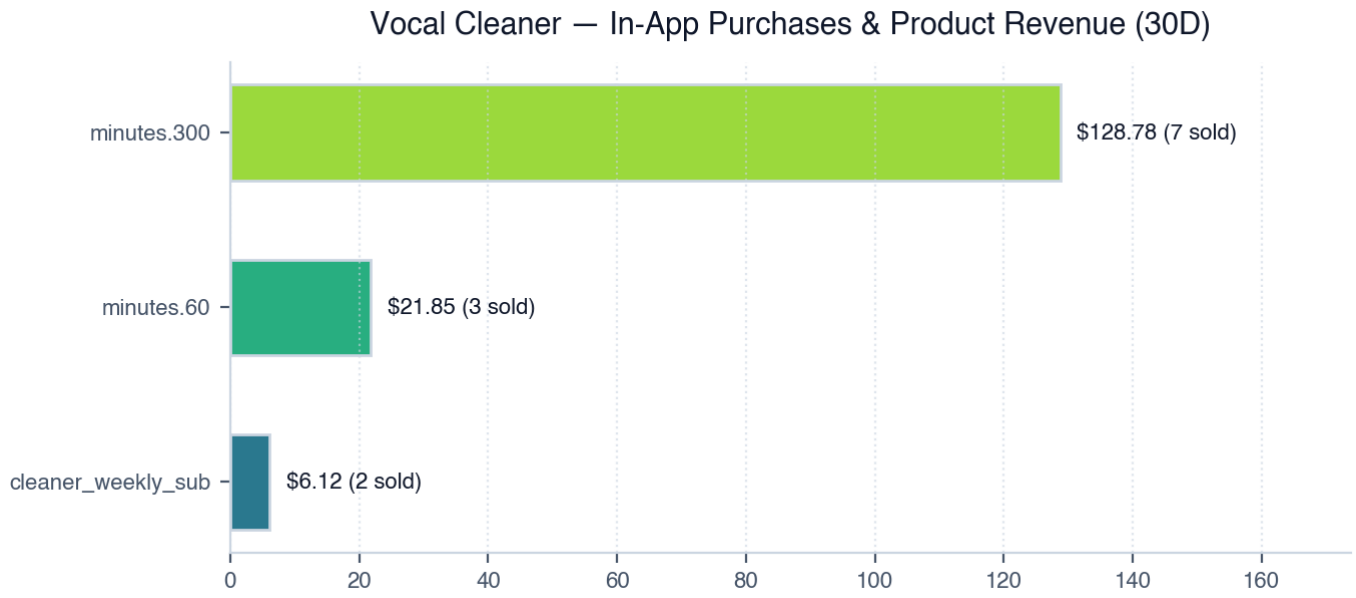
Music Separation Ios Version



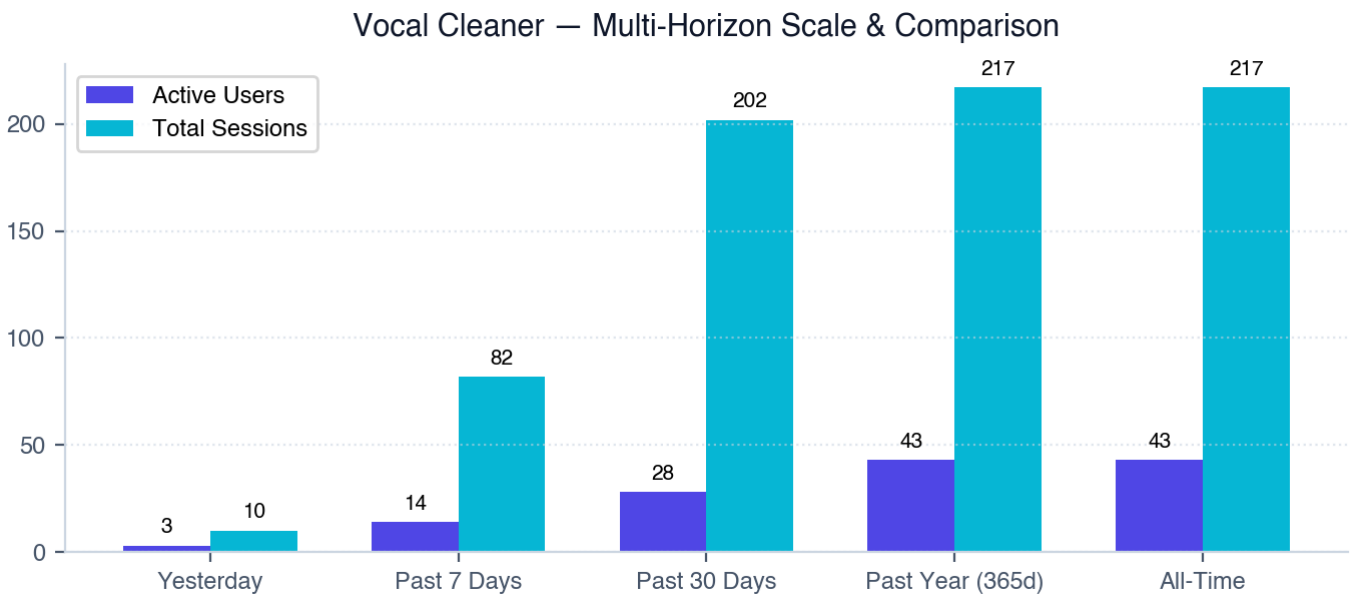
Vocal Cleaner Trend



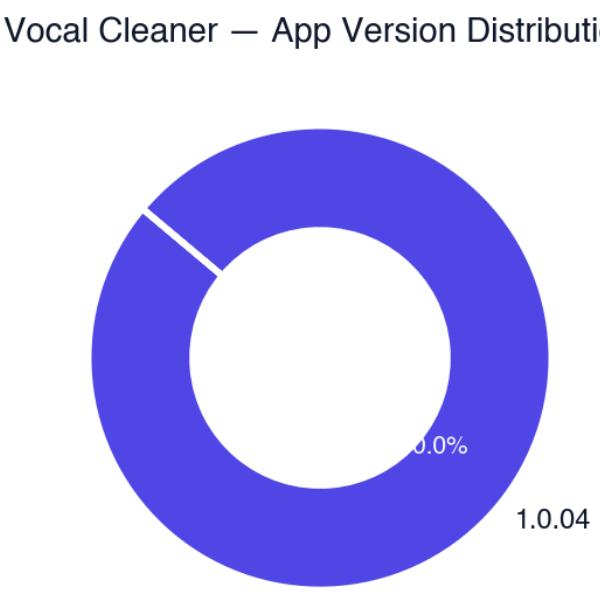
Vocal Cleaner Revenue



Vocal Cleaner Horizons



Vocal Cleaner Version



Direct Console Links:

■ Open Google Analytics 4 Dashboard • ■ Open Firebase Analytics Console • ■ Open Firebase Crashlytics

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